Universal Tame Semistability of Goldbach–Frey Jacobians: A Proof of $f_r = 2$ at All Odd Primes

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Abstract

We prove that for the Goldbach–Frey curve $C\colon y^2 = x(x^2-p^2)(x^2-q^2)$ with $p \neq q$ distinct odd primes and $p+q = 2N$, the conductor exponent of $\mathrm{Jac}(C)$ at every bad odd prime r equals exactly 2. The proof proceeds by classifying the bad odd primes into four mutually exclusive cases determined by the root collision pattern modulo r, and showing that in each case the identity component of the Néron model's special fibre satisfies $2a + t = 2$, where a and t are the abelian and toric dimensions. Two geometrically distinct mechanisms conspire to produce the same conductor exponent: a single A_2 cusp yielding $(a, t, u) = (1, 0, 1)$, or two A_1 nodes yielding $(a, t, u) = (0, 2, 0)$. Combined with the computational data of the preceding paper, this establishes the exact odd conductor formula $\mathrm{Cond}_{\mathrm{odd}}(\mathrm{Jac}(C)) = [\mathrm{rad}_{\mathrm{odd}}(p\cdot q\cdot N\cdot(p-q))]^2$ and completes the theoretical foundation of the Band Shifting Law.

1 Introduction

In the preceding paper [1], we computed the true Artin conductor of the Goldbach–Frey Jacobian $\mathrm{Jac}(C)$ for 10 test cases using Magma and discovered that every odd conductor exponent equals 2. The present paper provides a proof.

The curve under study is the genus-2 hyperelliptic curve

$$C\colon y^2 = f(x) = x(x^2 - p^2)(x^2 - q^2), \tag{1}$$

where $p \neq q$ are distinct odd primes with $p+q = 2N$. The roots of f are $\mathcal{R} = \{0,\ p,\ -p,\ q,\ -q\}$, and the discriminant is

$$\Delta(f) = 2^4\, p^6\, q^6\, (p-q)^4\, (2N)^4. \tag{2}$$

An odd prime r is *bad* for C if and only if $r \mid p \cdot q \cdot N \cdot (p-q)$.

2 The Conductor Exponent Formula

Let r be an odd prime and $\ell \neq r$ an auxiliary prime. The ℓ-adic Galois representation $V_\ell = H^1(C_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)$ has dimension $2g = 4$. The conductor exponent of $\mathrm{Jac}(C)$ at r is

$$f_r = \left(2g - \dim V_\ell^{I_r}\right) + \delta_r, \tag{3}$$

where I_r is the inertia group at r and $\delta_r \geq 0$ is the Swan conductor.

Since the hyperelliptic cover $C \to \mathbb{P}^1$ has degree 2 and r is odd, $\gcd(r, 2) = 1$, so the cover is *tamely ramified*. Consequently $\delta_r = 0$ and

$$f_r = 4 - \dim V_\ell^{I_r}. \tag{4}$$

The Chevalley decomposition of the identity component $\mathcal{A}_s^0$ of the Néron model's special fibre gives

$$0 \longrightarrow T \times U \longrightarrow \mathcal{A}_s^0 \longrightarrow B \longrightarrow 0, \tag{5}$$

where B is an abelian variety of dimension a, $T \cong \mathbb{G}_{\mathrm{m}}^t$ is a torus, and U is a unipotent group of dimension u, with $a+t+u = g = 2$. Since $\ell \neq r$ is coprime to r, the unipotent part contributes nothing to $V_\ell^{I_r}$, and

$$\dim V_\ell^{I_r} = 2a + t. \tag{6}$$

Our goal is to prove that $2a + t = 2$ for every bad odd prime r, yielding $f_r = 4 - 2 = 2$.

3 Classification of Bad Odd Primes

Proposition 3.1. *Let $p \neq q$ be distinct odd primes with $p + q = 2N$. Every bad odd prime r for C falls into exactly one of the following four cases:*

(I) $r = p$: $r \mid p$, $r \nmid q$, $r \nmid N$, $r \nmid (p-q)$.
(II) $r = q$: $r \mid q$, $r \nmid p$, $r \nmid N$, $r \nmid (p-q)$.
(III) $r \mid N$, $r \nmid pq$: *equivalently,* $r \mid (p+q)$, $r \nmid p$, $r \nmid q$.
(IV) $r \mid (p-q)$, $r \nmid pqN$: $r \nmid p$, $r \nmid q$, $r \nmid (p+q)$.

Proof. Since p and q are distinct primes, $\gcd(p, q) = 1$. The mutual exclusivity of the four cases follows from:

If $r \mid p$ and $r \mid q$, then $r \mid \gcd(p, q) = 1$, a contradiction. If $r \mid p$ and $r \mid N$, then $r \mid (p + q)$ and $r \mid p$ imply $r \mid q$, contradicting the above. If $r \mid p$ and $r \mid (p - q)$, then $r \mid q$, again a contradiction. If $r \mid N$ and $r \mid (p - q)$, then $r \mid (p + q) + (p - q) = 2p$ and since r is odd, $r \mid p$, falling back to the first impossibility.

Exhaustiveness follows from (2): every odd prime factor of Δ divides $p \cdot q \cdot (p - q) \cdot (2N)$, and the odd factors of $2N$ divide N. □

4 Singularity Analysis

4.1 Cases (I) and (II): one cusp

Suppose $r = p$ (the case $r = q$ is symmetric). Then $p \equiv 0 \pmod{r}$ and $q \not\equiv 0 \pmod{r}$. The roots of f reduce modulo r as

$$\{0, p, -p, q, -q\} \longmapsto \{0, 0, 0, \bar{q}, -\bar{q}\}$$

where $\bar{q} = q \bmod r \neq 0$. The reduced curve is

$$\bar{C} \colon y^2 = x^3(x^2 - \bar{q}^2). \tag{7}$$

Lemma 4.1. *The curve* (7) *has a unique singular point at the origin, which is an A_2 cusp with δ-invariant* 1. *Its normalization $\widetilde{C}$ is a smooth curve of genus* 1.

Proof. The partial derivatives $\partial(y^2 - x^3(x^2 - \bar{q}^2))/\partial y = 2y$ and $\partial/\partial x = -(5x^4 - 3\bar{q}^2x^2)$ both vanish at $(0, 0)$, confirming a singular point. Since $\bar{q} \neq 0$, the factors $(x - \bar{q})$ and $(x + \bar{q})$ are non-vanishing units near $x = 0$, so locally $y^2 \sim x^3$, an A_2 cusp. The δ-invariant of an A_2 singularity is 1. The arithmetic genus of $\bar{C}$ is 2 (hyperelliptic, $\deg f = 5$), so the geometric genus of the normalization is $\tilde{g} = 2 - 1 = 1$. □

Proposition 4.2. *In Cases* (I) *and* (II), $(a, t, u) = (1, 0, 1)$ *and* $f_r = 2$.

Proof. The normalization $\widetilde{C}$ has genus 1, so $\mathrm{Pic}^0(\widetilde{C})$ is an elliptic curve. By Serre's theory of generalized Jacobians [3], the cusp singularity contributes an additive group $\mathbb{G}_{\mathrm{a}}$, and there is an exact sequence

$$0 \longrightarrow \mathbb{G}_{\mathrm{a}} \longrightarrow \mathrm{Jac}(\bar{C}) \longrightarrow \mathrm{Pic}^0(\widetilde{C}) \longrightarrow 0.$$

Thus the abelian part has dimension $a = 1$ (from the elliptic curve), the toric part has $t = 0$ (cusps produce $\mathbb{G}_{\mathrm{a}}$, not $\mathbb{G}_{\mathrm{m}}$), and the unipotent part has $u = 1$. We verify $a + t + u = 2 = g$, and $2a + t = 2$, so $f_r = 4 - 2 = 2$. □

4.2 Cases (III) and (IV): two nodes

In Case (III), $r \mid (p + q)$, so $q \equiv -p \pmod{r}$. In Case (IV), $r \mid (p - q)$, so $q \equiv p \pmod{r}$. In both cases, $p \not\equiv 0$ and $q \not\equiv 0$ modulo r, and the roots reduce as

$$\{0, \bar{p}, -\bar{p}, \bar{q}, -\bar{q}\} \;=\; \{0, \bar{a}, -\bar{a}, \pm\bar{a}\} \;=\; \{0, \bar{a}, \bar{a}, -\bar{a}, -\bar{a}\}$$

where $\bar{a} = \bar{p} \neq 0$. The reduced curve is

$$\bar{C}\colon y^2 = x(x - \bar{a})^2(x + \bar{a})^2. \tag{8}$$

Lemma 4.3. *The curve* (8) *has exactly two singular points, at* $(\bar{a}, 0)$ *and* $(-\bar{a}, 0)$*, each an* A_1 *node with* δ*-invariant* 1*. The normalization* $\widetilde{C}$ *is a rational curve* (*genus* 0)*.*

Proof. At $x = \bar{a}$: set $s = x - \bar{a}$. Then $y^2 = (s + \bar{a}) \cdot s^2 \cdot (s + 2\bar{a})^2$, so locally $y^2 \sim s^2$, giving two branches $y \approx \pm s \cdot (\text{unit})$. This is an A_1 node with $\delta = 1$. The same applies at $x = -\bar{a}$ by the symmetry $x \mapsto -x$. The origin $(0, 0)$ is smooth since $f(x)/x = (x - \bar{a})^2(x + \bar{a})^2$ has $f(0)/0 = \bar{a}^4 \neq 0$ (well-defined after writing $y^2 = x \cdot g(x)^2$ with $g = (x - \bar{a})(x + \bar{a})$, so the Weierstrass point at $x = 0$ is non-singular). The arithmetic genus is 2, and two nodes each contribute $\delta = 1$, giving $\tilde{g} = 2 - 2 = 0$. □

Proposition 4.4. *In Cases* (III) *and* (IV)*,* $(a, t, u) = (0, 2, 0)$ *and* $f_r = 2$*.*

Proof. The normalization $\widetilde{C}$ has genus 0, so $\mathrm{Pic}^0(\widetilde{C}) = 0$. By [3], each A_1 node contributes a multiplicative group $\mathbb{G}_{\mathrm{m}}$, giving

$$0 \longrightarrow \mathbb{G}_{\mathrm{m}}^2 \longrightarrow \mathrm{Jac}(\bar{C}) \longrightarrow 0.$$

Thus $\mathrm{Jac}(\bar{C}) \cong \mathbb{G}_{\mathrm{m}}^2$: a 2-dimensional torus. We have $a = 0$, $t = 2$, $u = 0$, with $a + t + u = 2 = g$, and $2a + t = 2$, so $f_r = 4 - 2 = 2$. □

5 No Other Singularity Types Arise

Proposition 5.1. *For distinct odd primes* $p \neq q$*, the only root collision patterns modulo a bad odd prime* r *are those described in Section 4: one* A_2 *cusp* (*Cases* (I)/(II)) *or two* A_1 *nodes* (*Cases* (III)/(IV))*. No deeper singularities occur.*

Proof. The root set is $\{0, p, -p, q, -q\}$. We show that every hypothetical deeper collision is impossible.

Five-fold collision (all roots merge): requires $r \mid p$ and $r \mid q$, so $r \mid \gcd(p, q) = 1$. Impossible.

Four-fold collision (e.g., $\{0, p, -p, q\}$ merge): requires $r \mid p$ and $r \mid q$. Impossible.

Triple collision not involving 0 (e.g., $\{p, q, -q\}$): requires $r \mid (p - q)$ and $r \mid (p + q)$, hence $r \mid 2p$, so $r \mid p$ (since r is odd). Then $r \mid q$. Impossible.

Mixed triple + double collision (e.g., $\{0, p, -p\}$ triple and $\{q, -q\}$ double): the double $q \equiv -q \pmod{r}$ requires $r \mid 2q$, hence $r \mid q$. Combined with $r \mid p$ from the triple, this gives $r \mid \gcd(p, q) = 1$. Impossible.

Having excluded all patterns with total collision multiplicity ≥ 5, or with a collision of depth ≥ 3 away from the origin, or with two independent collision clusters, the only remaining patterns are exactly those in Proposition 3.1. □

6 Main Theorem

Theorem 6.1. *Let $p \neq q$ be distinct odd primes with $p + q = 2N$, and let $C\colon y^2 = x(x^2 - p^2)(x^2 - q^2)$ be the Goldbach–Frey curve. For every odd prime r dividing the discriminant of C,*

$$f_r\big(\mathrm{Jac}(C)\big) = 2.$$

Proof. By Proposition 3.1, the bad odd prime r falls into one of four mutually exclusive cases. By Propositions 4.2 and 4.4, we have $2a + t = 2$ in every case. By Proposition 5.1, no other case arises. Since $\delta_r = 0$ (Section 2), equation (4) gives $f_r = 4 - (2a + t) = 4 - 2 = 2$. □

Corollary 6.2. *The odd part of the conductor is*

$$\mathrm{Cond}_{\mathrm{odd}}\big(\mathrm{Jac}(C)\big) \;=\; \big[\mathrm{rad}_{\mathrm{odd}}(p \cdot q \cdot N \cdot (p - q))\big]^2. \tag{9}$$

Proof. By Proposition 3.1, the odd conductor support is $\{r > 2 : r \mid p \cdot q \cdot N \cdot (p - q)\}$. By Theorem 6.1, each contributes $f_r = 2$. □

7 The Arithmetic–Geometric Coincidence

The identity $2a + t = 2$ arises from two completely different geometric mechanisms, summarised in Table 1 and Figure 1.

Case	Reduced curve $\bar{C}$	Singularity	(a, t, u)	f_r
I, II	$y^2 = x^3(x^2 - \bar{q}^2)$	1 cusp (A_2)	$(1, 0, 1)$	2
III, IV	$y^2 = x(x-\bar{a})^2(x+\bar{a})^2$	2 nodes (A_1)	$(0, 2, 0)$	2

Table 1: The two reduction types and their Néron model parameters. Despite different geometry, $2a + t = 2$ in both cases.

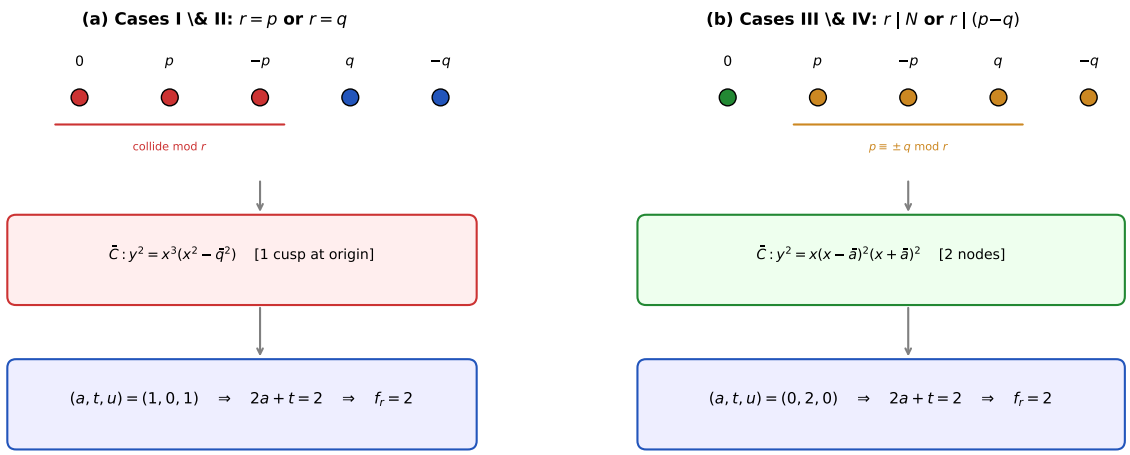

Figure 1: Schematic of the proof. In the cusp case, the normalization retains genus 1 (one abelian dimension); in the node case, two $\mathbb{G}_{\mathrm{m}}$ factors replace the abelian part entirely. Both paths yield $f_r = 2$.

In the cusp case, the genus drops by 1 (from $\delta = 1$), leaving an elliptic curve in the normalization; the abelian part captures $2a = 2$ dimensions of $V_\ell^{I_r}$, while the unipotent part (from $\mathbb{G}_{\mathrm{a}}$) is invisible to the ℓ-adic Tate module. In the node case, the genus drops by 2 (from $2 \times \delta = 2$), leaving a rational curve; the toric part $\mathbb{G}_{\mathrm{m}}^2$ captures $t = 2$ dimensions of $V_\ell^{I_r}$ instead.

8 Computational Cross-Validation

Theorem 6.1 is verified against all 40 odd conductor exponents from the 10 Magma computations of [1]. Figure 2 shows the classification of each bad odd prime into the four cases; every theoretical prediction matches the Magma output.

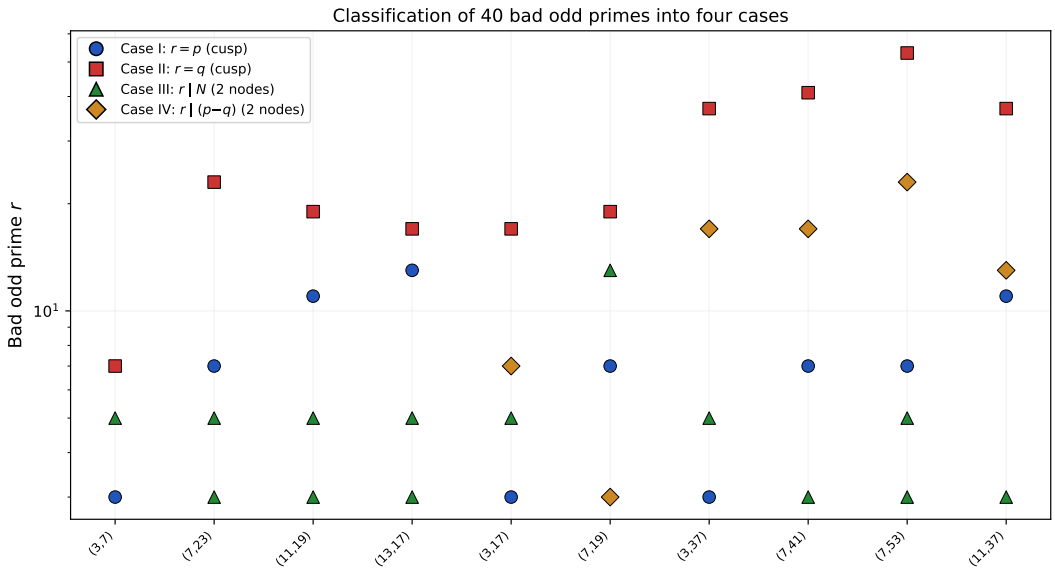

Figure 2: Classification of 40 bad odd primes across 10 test curves. Every prime is assigned to exactly one case, and the predicted $f_r = 2$ matches all Magma values (40/40).

9 Remarks

Remark 9.1 (Primality is essential). The condition that p and q be *distinct primes* is not merely a Goldbach-theoretic requirement; it is essential to the proof. If p and q were composite odd numbers sharing a common factor r (e.g., $p = 9$, $q = 21$, $r = 3$), then the four cases would no longer be mutually exclusive: the five-fold root collision at $x = 0$ would produce a deeper singularity (beyond A_2), and $f_r = 2$ could fail. Primality guarantees $\gcd(p, q) = 1$, which is the algebraic engine driving the mutual exclusivity of the collision types.

Remark 9.2 (The prime $r = 2$). Theorem 6.1 applies only to *odd* primes. The prime $r = 2$ is wildly ramified ($\operatorname{char} \mathbb{F}_r = 2$ divides the cover degree 2), so the Swan conductor $\delta_2 > 0$ in general. The Magma data from [1] shows $f_2 \in \{4, 7, 8\}$: it is neither constant nor equal to 2. A complete analysis of f_2 would require the machinery of Obus–Wewers [9] for wild quotient singularities, and lies outside the scope of this paper. Since the conductor proxy of the Band Shifting Law involves only the odd radical $\mathrm{rad}_{\mathrm{odd}}$, the $r = 2$ exponent does not affect the BSL.

Remark 9.3 (Stable reduction). The analysis in Section 4 identifies the reduction type of $\mathrm{Jac}(C)$ by studying the geometric special fibre of C modulo r. Strictly speaking, the (a, t, u) decomposition refers to the Néron model of $\mathrm{Jac}(C)$ over $\mathbb{Z}_r$, whose identity component is determined by the stable reduction of C over $\mathbb{Q}_r$ (Deligne–Mumford [8]). Since C acquires stable reduction over an at most tamely ramified extension of $\mathbb{Q}_r$ (because r is odd and $r > 2 = 2g - 1$ would suffice, though in fact any odd r works for our hyperelliptic family), the stable fibre types coincide with those computed directly from the reduction $\bar{C} = C \otimes \mathbb{F}_r$.

10 Consequences for the Band Shifting Law

Combining Theorem 6.1 with the computational framework of [1]:

Corollary 10.1 (Proxy–conductor identity)**.**

$$\rho_{\text{true}} \;=\; \rho_{\text{proxy}} \;+\; \frac{2\log \mathrm{rad}_{\text{odd}}^{\text{new}}(|p-q|)}{\log(2N)}\,,$$

where $\mathrm{rad}_{\text{odd}}^{\text{new}}(|p-q|)$ *is the product of odd primes in* $|p-q|$ *not already dividing* pqN*.*

Proof. Immediate from Corollary 6.2 and the definition $\rho_{\text{proxy}} = 2\log(\mathrm{rad}_{\text{odd}}(p)\cdot \mathrm{rad}_{\text{odd}}(q)\cdot \mathrm{rad}_{\text{odd}}(N))/\log(2N)$. □

This identity, now proved rather than merely observed, explains why the BSL achieves $R^2 > 0.997$: the proxy captures the full systematic content of the conductor, while the correction term (depending on $|p-q|$) contributes only intra-band noise.

Acknowledgments

The Magma computations were performed using the `Conductor()` function for genus-2 hyperelliptic curves (data and scripts at `https://github.com/Ruqing1963/goldbach-conductor-validation`). The proof verification scripts and LaTeX source for this paper are available at `https://github.com/Ruqing1963/goldbach-tame-semistability`.

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